

THUVAARAGAN THEVARAJAH

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Personal Statement

I am a seasoned AI/ML Engineer with 6 years of experience specializing in Generative AI, Computer Vision, AI Agents, Large Language Models (LLMs), NLP, Data Science, Machine Learning, and Deep Learning. I excel in rapid learning and problem-solving within dynamic teams. Passionate about using cutting-edge technologies, I deliver innovative AI solutions that drive measurable business impact. I thrive on challenges and effectively manage projects to advance AI innovations.

AI Engineer | *Exentai* | April 2025 – Present

End-to-End Development of an Agentic Workflow for a Modular AI Voice Agent

- Implemented an AI-powered agentic voice system enabling real-time user interaction via mobile phone calls.
- Developed modular agents (TTS, STT, LLM, Billing, Knowledge Base (Rag), Squad, MCP, Voicemail, and supporting tools) with emphasis on voice-to-text accuracy, reducing domain-specific transcription errors, and improving speaker diarization for high-quality transcriptions.
- Integrated diverse services: Twilio (telephony), ElevenLabs/Azure/Rime/Cartesia (TTS), Deepgram/Google/ElevenLabs (transcription), OpenAI/DeepSeek/Anthropic/Azure (LLM) and integrated LangChain for agent orchestration and LlamaIndex for efficient retrieval and knowledge integration.
- Engineered scalable cloud deployment using Python (OOP), FastAPI, PostgreSQL, Docker, and a robust CI/CD pipeline, ensuring maintainability, fault tolerance, and efficient production workflows.
- Deployed the ML pipeline on AWS EC2 with S3 for secure call-recording storage, generating automated end-of-call reports (summaries, scores, billing data). Used MLflow for experiment tracking and debugging to improve real-time reliability.

Hyper-Automated Financial Data Pipeline & Power BI Analytics

- Built an end-to-end hyper-automation pipeline to fetch financial data from Xero APIs for 30+ companies, replacing manual reporting workflows.
- Implemented automated ETL workflows using Python and Azure Power Automate, with scheduled execution on AWS Fargate, and ensured secure storage and monitoring using AWS S3 and CloudWatch.
- Delivered Power BI dashboards via Microsoft SharePoint integration, providing real-time financial insights and multi-company performance analytics.

AI/ML Engineer | 4 Axis Solutions (Pvt) Ltd | July 2023 – April 2025

Generate images from sketches

- Developed a complete pipeline for generating images from sketches based on text prompts.
- Deployed the pipeline on Google Cloud and created a Flask API to enable integration with an iOS device.
- Utilized advanced diffusion-based generative models and Hugging Face implementations including ControlNet, Mistoline, T2I Adapter, Stable Diffusion/SDXL, RealSRGAN, and MobileNet, while applying LLM and AI systems using Docker, PyTorch, TensorFlow, Python, and Swift. Tuned these models and analyzed sketch-variation limitations to optimize image generation quality.
- Improved memory efficiency and reduced generation time by 10%. This pipeline is live at SketchPro.com.

Recommendation system for drawing brushes

- Developed an AI-based feature for recommending brushes based on user prompts.
- Utilizing Stable Diffusion for texture generation and BERT-based word similarity for shape recommendation. Technologies used include PyTorch, LlamaIndex, NumPy, Pandas, scikit-learn, Python, and CI/CD pipeline.
- Developed an AI-based feature for recommending brushes by integrating VLM techniques (CLIP) to align text prompts with texture and shape features, enhancing personalization.

4 Axis Solutions - Chatbot Development

- Developed a RAG-based chatbot system leveraging LangChain & LlamaIndex for dynamic query responses.
- Integrated Pinecone vector database for efficient retrieval and semantic search. Utilized LLM(OpenAI API)(prompt engineering), NLP, Python, PyTorch, AI systems, CI/CD pipeline, and Google Cloud.

Developed Scalable AI Solutions for Inventory Optimization and Sales Forecasting

- Designed and optimized end-to-end data pipelines using **SQL, Python, Pandas, and NumPy** for stock analysis and sales forecasting, enhancing inventory management workflows and increasing demand prediction accuracy.
- Built advanced **time series forecasting** models using PCA, feature engineering, and ensemble techniques to deliver accurate, region-wise product forecasts for distributors.
- Developed scalable machine learning APIs and AI systems deployed on **cloud platforms (Google Cloud Vertex AI, AWS SageMaker, Azure)**, ensuring robust production performance and real-time business insights.
- Integrated interactive data visualization dashboards using **Matplotlib, and Power BI**, leveraging **PyTorch and NLP** techniques to support strategic decision-making and streamline operational workflows.
- Developed preprocessing pipelines including **missing-value imputation** and **multi-column constraint validation** to stabilize **model training** and improve **forecast accuracy**.

ML Associate Engineer | *SenzMate IoT Intelligence Solutions* | Jan 2022 – Dec 2022

Deep Learning-Based Crop Prediction

- I developed a system to **recommend suitable crops** to farmers based on **seasonal, regional, geo spatial, and IoT sensor data (such as soil moisture and temperature)**, and forecasted weather and crop prices to optimize agricultural planning.
- The system was built using **SQL, Python, scikit-learn, GeoPandas, Pandas, TensorFlow, PyTorch, Prophet, Neural Prophet, MLP, and LSTM**, with deployment pipelines managed using **Flask API, Docker, and CI/CD** for scalable integration.
- Gained experience in **data collection, handling missing data, outlier handling, feature engineering, one-hot encoding, data preprocessing, time series forecasting, data analytics, and data visualization** from this project.
- **Applied domain constraints** (weather, season, soil compatibility) and **diagnosed training issues from noisy IoT sensor data to improve model reliability**.

Healthcare Vision-Based Digital Physical Therapy

- The aim of this **healthcare** project was to analyze patient improvement during physiotherapy using video input of patients walking under specific conditions, with pose landmarks extracted via **Google MediaPipe and OpenCV**. **Spatial movement data** was processed using **Python and PyTorch** to build an AI system for accurate analysis.
- Developed a **Flask API** and containerized the CVML pipeline using **Docker**, deploying it on **AWS with CI/CD workflows** for reliable scaling and continuous updates.
- Developed a **CVML pipeline with an RTLS-style tracking system** to process pose landmarks, calculating walking speed, gait cycle phases, step length, stride length, and joint angles for accurate on-device assessment of patient improvement. This pipeline is live at [JOGO](#)

AI/ML Engineer | *Vital Masks (Pvt) Ltd.* | May 2019 – Dec 2021

Award-Winning Real Time Vision-Based Smart Parking System

- Developed a **Real-time spatial recognition and vision-based car slot detection system** to minimize parking time and enhance parking space management using automated slot availability tracking.
- Leveraged **real-time video processing, OpenCV, Python, C++, and federated learning to ensure privacy-preserving on-device inference, integrating SoTA CVML techniques in object detection and classification models optimized for edge deployment**, while incorporating **Generative AI model (GANs)** for synthetic data generation and augmentation to improve model robustness in real-world conditions.
- Secured **2nd Runner-Up position at TADHack 2022**, a global competition recognizing innovation in emerging tech solutions. [Vision-Based Smart Parking](#).

Research and Academic | *University of Moratuwa*

Award-Winning Spatial Image Captioning based blind Assistant system

- The **healthcare system** captures images in front of the blind and generates meaningful audible spatial-based captions to help them understand their surroundings.

- For this image captioning purpose, I chose Vision-Language Model (Vision Transformer as the image encoder and OpenAI's GPT-2 as the text decoder), implemented using PyTorch and Python, along with the Google synthesiser for TTS.
- Built and deployed a real-time ML pipeline on Raspberry Pi using TinyML, incorporating camera and ultrasonic sensors for edge device applications.
- This product was a finalist in the IEEE Sri Lanka-based Innovative Challenge 2023 and received an A+ grade for the final year project, which was a 10-credit module.

Personal Projects

Sentiment Analysis Using Deep Learning and LLMs

- This project implements sentiment analysis by classifying user queries as positive or negative using LSTM, python, PyTorch, DistilBERT, and LLaMA 3. It preprocesses, tokenizes, and splits the dataset into training and testing sets for optimal performance. The system was containerized and deployed on Azure Cloud for scalable inference. The project repository is available at GitHub - [Deep Learning-LLM](#).

Shop Bill Text Extraction Using Homography & Ocr

- This project utilizes Tesseract OCR and Homography to extract text from shop bills, correcting image distortions for accurate data retrieval. GitHub - [Homography](#).

Movie Genre Classification Using Fine-Tuned LLMs

- This project classifies movie genres based on a summary paragraph using fine-tuned LLMs, predicting one of five genres: action, family, romance, crime, or fantasy. The system leverages OpenAI's GPT-3.5 Turbo API and Meta's LLaMA-2-7B (prompt engineering) for accurate genre classification. GitHub - [Finetuning-LLM](#).

PDF Question Answering System with Graph Rag Integration

- This system extracts text from PDFs using Google Vision OCR, text processing by LlamaIndex, stores data in a Neo4j graph database, and processes user queries with OpenAI API (prompt engineering). It uses LangGraph to manage multi-step reasoning, handling queries across retrieval, LLM inference, and validation. It leverages graph relationships for accurate, context-aware question answering from PDF documents. The solution was hosted and orchestrated on Azure Cloud, ensuring secure storage and scalable performance.

Education

BSc. Eng.(Hons.) in Electronic and Telecommunication Engineering

Oct 2018 – May 2023

University of Moratuwa, Sri Lanka.

- CGPA – 3.69

G.C.E. Advanced Level Examination

2017

Jaffna Hindu College, Sri Lanka.

- Combined Mathematics, Physics and Chemistry 3A's
- Ranked 137th in the island (Z-score: 2.2532)

Technical Skills

Programming Languages : Python, Swift, SQL, C++

Machine Learning Models: Hugging Face Transformers, LLaMA(2 & 3), openai(GPT), BERT

AI Frameworks: TensorFlow, NumPy, PyTorch, OpenCV, Pandas, scikit-learn, Llama index, LangChain, vocode

Database: MongoDB, MySQL, PostgreSQL

Cloud Services: Google Cloud (Cloud Run, Cloud Functions, Vertex AI, and Cloud Storage), Salad Cloud, AWS (SageMaker, EC2, S3), Azure

DevOps: Docker, CI/CD, Git, Bitbucket, SourceTree

Visualization: Power BI, Matplotlib

Management Tools: Jira, Confluence

Hardware: Raspberry Pi, Arduino

Massive Open Online Courses (MOOC)

Machine Learning | Stanford | [Certificate](#)

Feb 2021

Neural Networks and Deep Learning | DeepLearning.AI | [Certificate](#)

Nov 2021

Introduction to Databases | Meta

Present

Generative AI with Large Language Models | DeepLearning.AI

Present

Honors and Awards

IEEE 5 MICC Finalist International Graph Signal Processing	April 2022
Codefest Datathon 2nd Runner Up National ML Competition	Jan 2023
TADHack 2nd Runner Up National Innovation Challenge	Oct 2022
Datastrom Semi-Finalist National ML Competition	Aug 2022
Mahapola for the Super Merit 137 th in Island National A/L Performance	Jan 2019

Clubs and Societies

Tamil Literary Association, UoM - Vice Secretary	Jan 2023 – Dec 2023
E-Club, UoM - Regular Member	2019 – 2023